

# Designing for deployment, and deciding whether an AI (artificial intelligence) project is worth it

Thursday 24 September 2026, 11:00 AM to 12:30 PM EDT (Eastern Daylight Time)

Videos V51 to V57, Module 10 (end) and Module 11

Running case: the McDonald's and IBM drive-thru voice ordering trial

## Student copy

The questions in this file are the ones asked live, without the answers. Work them through yourself, then check your reasoning against the session recording and the original decks. The hints are kept.

Questions during the session: Because of the interpretation delay, please send your questions in the Zoom chat, or better, after the session on the WhatsApp number you already have.

Block 1 of 7 · 11 minutes

## V51 — When neural networks win, and when they do not

A neural network learns its own features, and that only pays when the features are not already named.

## A neural network learns its own features, so it earns its cost on pictures, sound and free text, and rarely on a spreadsheet.

### How would a person describe a cat to a computer?

Pointed ears, whiskers, four legs, a tail. Then the light changes, the cat turns away, half of it is behind a chair, and the list stops working. One photograph is millions of pixel values and nobody can name which ones matter.

the word: features

### How would a person describe a late payer in a table?

The columns already say it: payment terms, days since the last order, region. A decision tree splits on payment terms in seconds and hands the credit controller a rule they can read out.

the word: tabular data

Three questions settle it before any model is picked: what is the data like, is there enough of it, and must each result be explained.

# The machine that was going to walk, talk and see, 1958

1

## A Navy press conference, July 1958

Frank Rosenblatt, a psychologist at the Cornell Aeronautical Laboratory, presents a machine that learns from examples instead of from rules written by hand.

2

## The next morning's newspaper

The New York Times reports that the Navy expects a machine that will walk, talk, see, write, reproduce itself and be conscious of its own existence. That claim came from the press conference, not from the laboratory.

3

## What was actually built

The Mark I Perceptron, shown publicly on 23 June 1960: 400 light sensors in a 20 by 20 grid, 512 middle units, 8 output units, weights set by small electric motors. It learned to tell simple shapes apart.

4

## 1969, the limits in print

Marvin Minsky and Seymour Papert published what this single-layer design could not learn. Interest and money moved elsewhere for years, and the idea only returned when machines and data caught up.

5

## What it leaves us

The machine now sits in the Smithsonian National Museum of American History. The idea was right and the description was far too large. Match the tool to the problem, and describe it accurately.

Checked against: Perceptron, Wikipedia — <https://en.wikipedia.org/wiki/Perceptron>

**1** A neural network works out its own features, which only helps when the features are not already named.

**2** On a table of named columns a simpler model is faster, cheaper and easier to defend.

**3** Three questions decide: the shape of the data, the amount of it, and whether a reason must be given.

A distribution company has 12,000 past orders in a table with 9 columns: customer size, region, product line, payment terms, days since the last order. They want to predict which invoices will be paid late. Their supplier proposes a neural network.

**What is the strongest reason to ask for something simpler?**

- A** Neural networks are always less accurate than simpler models.
- B** The columns are already named, so little is left to discover, and the credit team will ask why each invoice was flagged.
- C** Neural networks cannot handle dates.
- D** 12,000 rows is too much data for a neural network.

**Hint.** Ask what the network would have to work out that the table has already written down.

**In the session: type the letter in the Zoom chat.**

A bank receives 40,000 scanned loan files a month as images and wants to pull out the page that carries the customer signature. The layouts differ between branches and between years.

**Which approach fits, and why?**

- A** Write one rule by hand for each known layout.
- B** A neural network, because the input is an image where the useful patterns cannot be listed in advance.
- C** A decision tree on the file size of each page.
- D** The task cannot be automated.

**Hint.** Go back to the first of the three questions and name the kind of data.

**In the session: type the letter in the Zoom chat.**

An insurer has 600 labelled examples of a rare claim type. The regulator requires a specific reason whenever a claim is refused.

**What should the insurer do?**

- A** Build the neural network anyway and add an explanation tool afterwards.
- B** Choose a simpler model: 600 examples is far below what a network needs, and reasons are required.
- C** Collect 500,000 more examples first, then decide.
- D** Use the network for speed and keep the reasoning internal.

**Hint.** Two of the three questions answer no here. How many noes do you need?

**In the session: type the letter in the Zoom chat.**

## Run the three-question test

### What the numbers are

The three questions are the ones in the video, in the same order. Nothing here is generated at random: the same three settings always give the same verdict.

### What to move

Three settings, one for each question. The verdict and the reason are written out as you change them. Controls: 1. What is the data like?; 2. How many examples?; 3. Must each result be explained?.

### Try, in this order

1. Leave it as it opens: a table, a few hundred examples, reasons required. Read the verdict.
2. Change only the data to photographs. The verdict does not move yet; find out which question is still blocking it.
3. Now raise the amount to millions. One question is still saying no.

**The point.** One no is enough to rule a neural network out. All three have to say yes before it is the right answer.

- 1** Data shape decides first: pictures, sound and text point one way, rows and columns point the other.
- 2** Volume and computing power decide second, and a few hundred examples is not enough for a network.
- 3** If a reason must be given, that requirement outranks the accuracy figure.

Block 2 of 7 · 11 minutes

## V52 — Human-in-the-loop design

Decide where a person must approve before the system is built, not after the first complaint.

**Three facts place the person: can the decision be undone, what does a mistake cost, and how often is the model wrong on inputs like this one.**

**When must a person approve before the decision takes effect?**

When it is hard to undo, expensive when wrong, and wrong often enough to matter. Refusing a loan, dismissing a member of staff, approving a payment. Automating these first is the mistake.

**the word: reversibility**

**When is checking the record afterwards enough?**

When it is easy to undo, cheap, and usually right. A product recommendation, an email filed in the wrong folder. Review a sample of the log and move on.

**the word: confidence routing**

The drive-thru trial sat at the cheap end of that scale, and roughly one order in five still needed the person at the window.

# The night the warning system was confident and wrong, 1983

1

## Just after midnight, 26 September 1983

Lieutenant Colonel Stanislav Petrov is the duty officer at Serpukhov-15, the command centre for the Oko satellite early-warning system, on a shift he was not originally scheduled to work.

2

## The alarm

The system reports a missile launched from the United States, then a total of five. It rates its own confidence at maximum. Standing procedure is to pass the report straight up the chain of command.

3

## A few minutes to decide

Petrov reasons that a real first strike would not be five missiles, and that the ground radar has confirmed nothing. He reports the alert as a system fault.

4

## What the satellite had seen

There were no missiles. Sunlight reflected from high cloud tops produced a heat signature the sensors read as rocket exhaust. The design could not tell the two apart in that geometry.

5

## What it leaves us

He was reprimanded over his logbook and the incident stayed secret until 1998. The system was confident and wrong, and the only reason it did not act was that a person stood between the output and the decision.

Checked against: Stanislav Petrov, Wikipedia — [https://en.wikipedia.org/wiki/Staniislav\\_Petrov](https://en.wikipedia.org/wiki/Staniislav_Petrov)

**1**

Reversibility, cost of error and error rate together decide where the person sits.

**2**

Confidence routing sends the doubtful cases to a person and lets the confident ones through.

**3**

Place the people first and choose the model second, or you will fit controls to a finished model.

The returns model in the video catches 60 percent of risky returns. Specialists review the borderline 20 percent of orders and the rest are approved automatically. A manager proposes removing the human review to save the specialists' time.

**What is the strongest objection?**

- A The model would become less accurate without people looking at it.
- B A wrongly refused return is hard for the customer to undo and expensive in goodwill, so the three factors still point to review.
- C Human review is required by law in all cases.
- D The model stops working if nobody watches it.

**Hint.** Run the three factors over one wrongly refused return, from the customer's side.

**In the session: type the letter in the Zoom chat.**

A returns team handles 1,000 orders a day. At the current confidence cut-off the model sends 200 of them to a person. Each review takes 4 minutes and a reviewer costs 30 dollars an hour. Finance asks what happens if the cut-off is moved so that only 80 orders a day are reviewed.

**What happens to the cost and to the mistakes?**

- A** Review cost falls from about 400 dollars a day to about 160, and more of the model's mistakes go through unchecked.
- B** Review cost falls and the model's accuracy improves.
- C** Nothing changes; the model still makes the same decisions.
- D** Review cost rises, because fewer reviewers work more slowly.

**Hint.** 200 orders at 4 minutes is 13.3 hours. Work out the same figure for 80.

**In the session: type the letter in the Zoom chat.**

In the drive-thru trial, order accuracy was reported in the low 80 percent range against a target above 95 percent, and one reported difficulty was handling different accents.

**Where does a wrong order sit on the three factors?**

- A** Hard to undo, expensive, rare: a person must approve every order.
- B** Easy to undo, cheap on its own, but happening often: keep a person present to correct, and do not expand until the error rate falls.
- C** Easy to undo, cheap, rare: full automation is fine.
- D** The three factors do not apply to voice systems.

**Hint.** Two of the three factors are mild here. The third is not.

**In the session: type the letter in the Zoom chat.**

## Move the confidence cut-off

### What the numbers are

The 1,000 orders are generated once from a fixed starting value, so the same cut-off always gives the same numbers. The model is wrong on about 12 percent of orders overall, and far more often on the ones it is least sure about.

### What to move

1,000 orders a day. The model is sure about most of them and unsure about the rest, and the reviewers see the least confident ones first. Controls: Orders sent to a person; Minutes per review; Cost of an hour of review.

### Try, in this order

1. Leave the cut-off at 200 and write down the daily cost and the mistakes that escape.
2. Drop it to 80. The cost falls by 240 dollars a day; count how many more mistakes reach the customer.
3. Now find the cut-off where nothing escapes, and read what that costs.

**The point.** There is no setting with no cost. You are choosing between review hours and escaped mistakes, and the right point depends on what one escaped mistake costs you.

**1** Ask the three questions about the decision, not about the model.

**2** Routing by confidence puts the review effort where the doubt is.

**3** Design the checkpoint before choosing the model, or you will be retrofitting controls.

Block 3 of 7 · 12 minutes

## V53 — The data availability audit

Sort every field into available now, obtainable with effort, or too costly, and write it on one page.

**The audit is run before anyone is assigned to the project, because its job is to decide whether the project can be built at all.**

**What happens when nobody asks what data exists?**

Six months in, the licence fee blocks the timeline, or the field was never collected, or it lives in three systems that do not speak to each other. The team is discouraged, the project is shelved, and nobody writes down why.

**the word: data silo**

**What does the audit produce?**

Every candidate field sorted into available now, obtainable with effort, or too costly. Then one page saying what version one contains, what comes later, and what is deliberately left out.

**the word: data readiness statement**

Missing data is an engineering problem. Expensive data is a business decision. Keeping the two apart is most of the value of the audit.

## Two systems, two sets of units, one lost spacecraft, 1999

1

### 11 December 1998, launch

NASA launches the Mars Climate Orbiter. With its companion lander the programme costs about 327 million dollars. It is meant to study the Martian atmosphere and relay data from the surface.

2

### The join between two teams

During the cruise, small thruster firings keep the craft on course. Ground software from one contractor reports them in pound-force seconds. The navigation software on the other side expects newton-seconds.

3

### Nobody checks the boundary

Each system is correct within itself. Neither team verifies the units where the two meet, and every small correction is out by a factor of about four and a half.

4

### 23 September 1999

The craft arrives at Mars far lower than planned and is lost, almost certainly destroyed in the atmosphere. Nine months of travel end at a data boundary that nobody owned.

5

### What it leaves us

This is the first obstacle in this video in its purest form. Data in separate systems is not simply data in two places. It is data in two languages, and the join is where projects fail.

Checked against: Mars Climate Orbiter lost, Physics Today — <https://physicstoday.aip.org/news/mars-climate-orbiter-lost>

**1** Ask what exists, what is obtainable and what is too costly, before anyone is assigned to the work.

**2** A stand-in field that is already collected often beats the ideal field that is not.

**3** The output is one page, shared with everyone, and updated when circumstances change.

The churn audit in the video has eight candidate fields. Product usage logs exist in the analytics system, but the customer identifiers do not match the main customer system, so joining them needs a four-week piece of work. Bought social media sentiment would cost 100,000 dollars a year plus two to three months of legal review.

**Which bucket does each one belong in?**

- A Both belong in too costly right now.
- B Usage logs are obtainable with effort and belong in version two; the bought sentiment is too costly and is named as excluded.
- C Both are obtainable with effort, so both go into version one.
- D Neither belongs in the audit because neither is available today.

**Hint.** One of them needs work you can schedule. The other needs a budget conversation and a contract.

**In the session: type the letter in the Zoom chat.**

The team cannot get exact dates on which customers stopped being active. Someone proposes using the gap between the last transaction and today instead.

**What should the team do?**

- A** Refuse: a stand-in field is never acceptable in a serious model.
- B** Accept it as a stand-in, note it in the readiness statement, and check afterwards whether the model still performs.
- C** Accept it and record it as the exact churn date.
- D** Delay the project until the exact dates are collected.

**Hint.** The question is not whether it is perfect, but whether it is honest and whether it works.

**In the session: type the letter in the Zoom chat.**

A project is six months in when the team discovers that the data licence blocking their timeline costs more than the remaining budget. The work is quietly shelved and no report is written.

**What went wrong, and when?**

- A** Bad luck: nothing in the audit would have prevented it.
- B** The third obstacle in the video: an expensive source that was never named, so it stopped the project late instead of early.
- C** The team should have bought the licence at the start regardless of cost.
- D** The team should have used a neural network instead.

**Hint.** Ask when the cost became known, not whether it was affordable.

**In the session: type the letter in the Zoom chat.**

## Run the audit on eight fields

### What the numbers are

The eight fields, the four-week identity-matching sprint and the 100,000 dollar licence all come from the video. Demographics, tenure, account value, transaction history and support tickets are fixed as available now.

### What to move

Five fields are already in the company's own systems. You decide where the other three go, and the readiness statement rewrites itself. Controls: Product feature usage logs; Customer reviews; Bought social media sentiment.

### Try, in this order

1. Leave it as the video has it and read the statement underneath.
2. Move the bought sentiment into version one, and watch the start date and the yearly cost.
3. Now move the usage logs into version one as well, and see how much later version one can begin.

**The point.** The buckets are not opinions. Each one carries a date and a cost, and moving a field between them moves the project.

**1** Run the audit before the project has a team and a timeline, not after.

**2** Keep missing data and expensive data in separate columns; they need different fixes.

**3** One page, shared, and revisited when something changes.

Block 4 of 7 · 10 minutes

## V54 — When accuracy is not enough

A model nobody will act on produces nothing, whatever its score says.

## Decide the level of explanation the use case requires first, and let that rule out the models which cannot meet it.

### What did the 91 percent model deliver?

Nothing. The hospital's ensemble was right more often than anything else available, and the clinicians could not see what it was reacting to, so they did not act on it. A model nobody follows has a business value of zero.

**the word: black box**

### What did the 88 percent model deliver?

Nearly all of its potential. A decision tree can be read aloud, so it was used, and three points of accuracy bought a large rise in the number of patients actually helped.

**the word: interpretability**

Post-hoc tools help where the model is already understandable. They do not turn an opaque one into a compliant one.

## The right answer that nobody would accept, Vienna 1847

1

### Two clinics, side by side

Vienna General Hospital runs two free maternity clinics and admits women by the day of the week. In the clinic staffed by doctors and students about one mother in ten dies of childbed fever. Next door, with the midwives, it is under one in twenty-five.

2

### What Semmelweis noticed

The doctors come to the ward straight from post-mortem examinations and the midwives do not. In May 1847 he orders hands to be washed in a chlorinated lime solution before examinations.

3

### The result

Deaths in his clinic fall to roughly one in a hundred. In 1848, the first full year of the rule, the doctors' clinic is safer than the midwives' clinic next door.

4

### The profession refuses

He cannot say what the washing removes, because the germ theory of disease is still twenty years away. Colleagues are offended by the suggestion that their hands carry anything, and the practice is not adopted.

5

### What it leaves us

He was right, the evidence was strong, and without an explanation people would accept it changed almost nothing in his lifetime. He died in 1865. Being correct is not the same as being adopted.

Checked against: Ignaz Semmelweis, Wikipedia — [https://en.wikipedia.org/wiki/Ignaz\\_Semmelweis](https://en.wikipedia.org/wiki/Ignaz_Semmelweis)

**1** Set the level of explanation the use case requires before the shortlist of models, not after.

**2** Tools that estimate what drove a prediction help where the model is already understandable.

**3** An unused model has a business value of zero, whatever the test set says.

A hospital has two models for readmission risk. The ensemble reaches 91 percent accuracy and the clinicians will not act on it. A decision tree reaches 88 percent, and the clinicians use it.

**Which one should be deployed?**

- A** The 91 percent model, because accuracy is the objective measure.
- B** The 88 percent model, because three points of accuracy are worth less than the difference between being used and being ignored.
- C** Both, and let each clinician choose.
- D** Neither, until one of them reaches 95 percent.

**Hint.** Multiply each model's accuracy by the share of cases where anyone acts on it.

**In the session: type the letter in the Zoom chat.**

A lender is told by its legal team that a refused applicant must be given the specific reasons for the refusal. The data science team proposes keeping the complex model and adding a tool that estimates which inputs drove each prediction.

**Is that enough?**

- A** Yes: the tool produces reasons, which is what the rule asks for.
- B** Not on its own: an estimate of what drove an opaque model is not the model's reasoning, and the rule is about the decision.
- C** Yes, provided the tool is bought from a large supplier.
- D** The rule does not apply to models, only to human decisions.

**Hint.** Ask what the tool is explaining: the decision, or its own approximation of the decision.

**In the session: type the letter in the Zoom chat.**

A retailer's fraud team is offered a complex model that catches slightly more fraud than the current rule-based system. The security and compliance teams say they cannot check what it is doing.

**How much weight should that objection carry?**

- A** None: fraud caught is the only measure that matters.
- B** A lot: the checkable model clears approval faster, so it starts saving money sooner and produces fewer internal escalations.
- C** None, if the supplier keeps the method confidential so there is nothing to argue about.
- D** None: run the complex model and inform compliance later.

**Hint.** Count the time between choosing a model and having it actually running.

**In the session: type the letter in the Zoom chat.**

## Accuracy against adoption

### What the numbers are

The two models are the ones in the video: 91 percent from the ensemble, 88 percent from the decision tree, and a large gap in how often clinicians act. The adoption figures are placed at 20 and 90 percent to show the size of the effect.

### What to move

The hospital's two models, side by side, with the number of patients each one actually helps in a year. Controls: Model in production; Explanation tool bolted on; High-risk patients flagged a year.

### Try, in this order

1. Start with the decision tree and write down the patients helped.
2. Switch to the ensemble. Accuracy rises three points; see what happens to patients helped.
3. Now bolt the explanation tool onto the ensemble and check whether it closes the gap.

**The point.** Accuracy is multiplied by adoption before it becomes a result. A small gain in the first is easily wiped out by a loss in the second.

**1** Name the explanation requirement before the model shortlist.

**2** Tools added afterwards do not convert an opaque model into a compliant one.

**3** Accuracy counts only once it is multiplied by the share of cases where someone acts.

Block 5 of 7 · 12 minutes

## V55 — The real cost of deploying AI models

Building the model is the one-off cost; keeping it alive is the one that repeats every year.

## Four costs start on the day the model goes live: infrastructure, monitoring, retraining and the people who run all three.

### What is on the approval slide?

Fifty thousand dollars to build the churn model, and 95 percent accuracy. This is the figure the project is approved on, and it is the smaller of the two figures that matter.

the word: training cost

### What is on next quarter's invoice?

Thirty thousand for infrastructure, forty for monitoring, twenty for retraining and two hundred for one and a half engineers. That is 290,000 dollars a year, every year the model runs.

the word: cost of ownership

At 5,000 dollars of lifetime value per customer kept, the model must save 58 customers a year before it has paid for itself.

# The bridge that was never finished being painted, 1890 to 2011

1

**1883 to 1890, the build**

The Forth Bridge is built across the Firth of Forth in Scotland: a steel railway bridge held together by about six and a half million rivets. It opens in 1890 and the construction bill is settled.

2

**Then the salt air**

Steel in a sea estuary corrodes, so the structure has to be painted. Crews work along it year after year, and the phrase for a job that never ends enters the language.

3

**The cost nobody planned for**

The build was one project with an end date. The upkeep is a standing team, repeating indefinitely, and across a century it costs far more than the original construction did.

4

**2002 to 2011, doing it properly**

A ten-year programme costing about 130 million pounds strips the structure back to bare metal and recoats 230,000 square metres, with up to 400 people a day on the bridge at the peak.

5

**9 December 2011**

The scaffolding comes down and the new coating is expected to last about 25 years. The thing you build is not the expensive part, and the maintenance schedule is set by the weather, not by the calendar you would prefer.

Checked against: Network Rail, Forth Bridge repainting complete — <https://www.networkrailmediacentre.co.uk/news/finished-on-the-forth-bridge-repainting-now-complete>

- 1** Four recurring costs begin at deployment: infrastructure, monitoring, retraining and labour.
- 2** Labour is normally the largest of the four, and it repeats for as long as the model is live.
- 3** Retraining rhythm is set by how fast your world changes, measured, not by a round number on a calendar.

The churn model costs 50,000 dollars to build and 290,000 dollars a year to run. A director asks what the board should be shown before approving it, assuming the model runs for five years.

**Which figure belongs in the approval paper?**

- A** The 50,000 dollar build cost and the accuracy figure.
- B** About 1.5 million dollars: the build plus five years of running costs, set against what the model should save over the same five years.
- C** 290,000 dollars, because that is the recurring cost.
- D** Nothing, until the model has been running for a year.

**Hint.** Fifty thousand once, plus 290,000 multiplied by five.

**In the session: type the letter in the Zoom chat.**

A fraud model trained two years ago starts meeting payment types that did not exist when it was built. Its accuracy slips and nobody notices for four months.

**What is happening, and what was missing?**

- A** Concept drift: the relationship between inputs and outcome has changed.
- B** Data drift: the incoming data now contains things the model has never seen, while the relationship is unchanged. Monitoring is what was missing.
- C** A bug in the model code.
- D** Normal behaviour: no action is needed.

**Hint.** Ask whether the world's rules changed, or only the traffic arriving at the door.

**In the session: type the letter in the Zoom chat.**

To reduce cost, a company staffs its production model with half an engineer instead of one and a half. Two years later the model is unmaintainable, incidents are frequent, and the project restarts from the beginning.

**What did the company actually save?**

- A Two years of salary, so the decision was reasonable.
- B Nothing: monitoring improvements and retraining were skipped, and the restart costs more than the salaries avoided.
- C Nothing, because the model should have been more accurate.
- D Nothing, because they should have bought a bigger server.

**Hint.** Ask what work stopped happening, and what it cost to do that work later.

**In the session: type the letter in the Zoom chat.**

## The five-year cost of ownership

### What the numbers are

Every starting figure comes from the video. An engineer is costed at 133,000 dollars a year, so one and a half of them is the 200,000 in the video's own example.

### What to move

The churn model from the video. Infrastructure is 30,000 and retraining 20,000 a year; the build cost 50,000 once; one customer kept is worth 5,000. Controls: Engineers, full time; Monitoring; Years the model will run.

### Try, in this order

1. Leave everything as it opens and confirm the two headline numbers: 290,000 a year and 58 customers.
2. Set monitoring to none. The break-even figure falls; read what happens to the months before a problem is noticed.
3. Put monitoring back and raise the engineers to 2.5. Read the five-year total.

**The point.** Labour moves the total more than anything else on the screen, and the line that is easiest to cut is the one that tells you when the model has stopped working.

**1** Show the five-year cost of ownership in the first approval conversation, not the build cost.

**2** Labour is usually the largest recurring line, and it does not end.

**3** Monitoring is what turns a silent decline into something you can act on.

Block 6 of 7 · 12 minutes

## V56 — Linking model performance to business value

Translate the measure into revenue, cost or risk, or nobody can decide anything with it.

**Every business benefit arrives through one of three channels, and a model's performance means nothing until it has been turned into one of them.**

**What does 92 percent accuracy tell the finance director?**

Nothing they can act on. Accuracy, precision and ranking scores are not revenue, cost or risk, and the question on the table is whether to spend 500,000 dollars on deployment.

**the word: accuracy**

**What do three percentage points of default rate tell them?**

Eight percent down to five percent on a 100 million dollar portfolio is 3 million dollars of expected losses prevented, set against 50,000 dollars to build. That is a return of about sixty to one.

**the word: expected loss**

A rough estimate with money attached beats a precise measure with none, because it frames the real question: what performance is good enough to break even.

# The map that was worth exactly one pump handle, London 1854

1

## 31 August 1854

A cholera outbreak begins in a few streets of Soho. About 500 people die within ten days. The accepted explanation at the time is bad air.

2

## The doctor with a different idea

John Snow believes the cause is the water. He obtains the register of deaths, walks the streets to fill the gaps, and marks every death on a map of the neighbourhood.

3

## What the map showed

The marks cluster around one public pump on Broad Street. Two oddities strengthen the case: the workhouse with its own well is barely touched, and a woman who died in Hampstead had that water delivered because she preferred the taste.

4

## 7 and 8 September

Snow puts the evidence to the parish board. They are sceptical, and they agree. The next morning the handle is taken off the pump so the well cannot be used.

5

## What it leaves us

Snow was careful in print: deaths were already falling because people had fled. The map's value was never the picture. It was that it changed one decision, and acting on it cost one pump handle.

Checked against: London Museum, John Snow and the Broad Street pump — <https://www.londonmuseum.org.uk/collections/london-stories/john-snow-cholera-broad-street-pump/>

- 1** Revenue, cost or risk: if a project fits none of the three, the case is not yet made.
- 2** Expected loss is the chance of the bad outcome multiplied by what it costs, which turns uncertainty into a number people can argue about.
- 3** Model performance only matters when it changes a decision somebody was going to make anyway.

A credit team's old model put the default rate on a 100 million dollar portfolio at 8 percent. The new model puts it at 5 percent. Building and deploying it costs 50,000 dollars.

**What is the value of the new model?**

- A** The 3 percentage point improvement in accuracy.
- B** Three percent of 100 million dollars, which is 3 million dollars of expected losses prevented, against a 50,000 dollar cost.
- C** It cannot be estimated until a year has passed.
- D** One hundred million dollars.

**Hint.** Take the difference between the two rates and apply it to the portfolio.

**In the session: type the letter in the Zoom chat.**

A fraud model reduces false alarms by 20 percent. The team currently spends 1,000 hours a month reviewing flagged transactions, at a fully loaded cost of 75 dollars an hour.

**What is the saving?**

- A** 200 hours a month, which is 15,000 dollars a month or 180,000 dollars a year.
- B** It cannot be calculated without knowing the model's accuracy.
- C** Twenty percent of the fraud losses.
- D** Nothing, because the reviewers are paid a salary either way.

**Hint.** Twenty percent of 1,000 hours, then multiply by the hourly cost.

**In the session: type the letter in the Zoom chat.**

The technical team's business case assumes the retention campaign converts at 40 percent. The marketing director says the realistic figure is 25 percent. The model is already built.

**What should happen next?**

- A Proceed with 40 percent, because the technical team built the model.
- B Settle the assumption now, because the whole value estimate rests on it; if it cannot be settled, run a small pilot and measure it.
- C Average the two figures and use 32.5 percent.
- D Drop the project, because the disagreement proves the model does not work.

**Hint.** Ask which number the value estimate is actually built on.

**In the session: type the letter in the Zoom chat.**

## Turn the measure into money

### What the numbers are

Each channel uses the video's own figures at full delivery: 3 percentage points off an 8 percent default rate on a 100 million dollar portfolio, 200 review hours a month at 75 dollars, and 58 customers kept at 5,000 each.

### What to move

The same model, translated through each of the three channels in turn, against what it costs to build. Controls: Which channel?; How much of the improvement is delivered; Build and deployment cost.

### Try, in this order

1. Stay on risk reduction at full delivery and read the value and the return.
2. Pull delivery down to 40 percent. Find where the return drops below ten to one.
3. Switch to cost reduction at the same delivery, and compare the two returns.

**The point.** The same model looks decisive or marginal depending on which channel you translate it into, so choose the channel before you build the case.

**1** Pick the primary channel first: revenue, cost or risk.

**2** Write the case as one sentence: if the model reaches this, then this happens, therefore this much money.

**3** Name the assumptions out loud, because that is where the case usually fails.

Block 7 of 7 · 12 minutes

## V57 — Organizational readiness for machine learning

Two pillars must stand together, and the weaker one decides whether you build, buy or partner.

**Readiness is not a score out of ten. It is a diagnosis of which pillar is weak, and what that costs you in months.**

**What does a company with perfect data and no specialists get?**

Eighteen months of work produced a clean central data store and excellent data engineers, and nobody who could build or maintain a model. The bought-in models arrived and no one could explain them.

**the word: data infrastructure**

**What does a company with four specialists and scattered data get?**

Four highly qualified people waiting. Customer data sits in three systems that do not talk to each other, joining them takes eight months, and until then there is nothing to work on.

**the word: team capability**

Neither company was unready in general. Each was unready in one specific, nameable way, and that name is what decides the path.

## The tower that was saved by a hundred-year delay, Pisa 1173

**1**

**9 August 1173**

The foundations of the bell tower at Pisa are laid: about three metres deep, in soft and unstable ground. Everything above assumes the ground will behave.

**2**

**1178, two or three floors up**

One side is sinking and the tower is visibly leaning. Work stops. Nothing about the masonry above is at fault; the constraint is entirely underneath it.

**3**

**Nearly a century of war**

Pisa is repeatedly at war and the site is left alone for close to a hundred years. The pause lets the ground settle, and it is most likely the reason the tower did not fall.

**4**

**1272 to 1372**

Building resumes. The upper floors are made taller on one side to compensate for the tilt, which is why the tower is not only leaning but slightly curved. The bell chamber is finished about 199 years after the first stone.

**5**

**1990 to 2001**

Closed at a lean of 5.5 degrees and judged at risk, it is stabilised by removing soil from under the high side and reopened in December 2001 at about 3.97 degrees. The ground set the limit for eight centuries, which is what a weak pillar does.

Checked against: Institution of Civil Engineers, Stabilising the Leaning Tower of Pisa — <https://www.ice.org.uk/what-is-civil-engineering/what-do-civil-engineers-do/stabilising-the-leaning-tower-of-pisa/>

**1** Both pillars have to stand at the same time; strength in one does not compensate for weakness in the other.

**2** Match the path to the gap, not to the budget: build, buy and partner each demand different things.

**3** A gap is only useful once it has a name, a timeline and an owner.

An insurer has decades of clean claims data, a working data platform, and no machine learning staff at all. Underwriting matters to them, but it is not what distinguishes them from competitors.

**Which path fits?**

- A** Build in-house, because they have the data.
- B** Buy or partner, because the data pillar is strong, the team pillar is empty, and hiring a full team first would cost them a year.
- C** Hire five specialists and then decide.
- D** Do nothing until both pillars are strong.

**Hint.** Name which pillar is missing, then ask which path fills that particular gap fastest.

**In the session: type the letter in the Zoom chat.**

A second insurer has two capable machine learning engineers. Claims data sits in legacy systems from three past mergers with no unified way to reach it. The head of analytics proposes hiring a third engineer.

**What should they do instead?**

- A Hire the third engineer, because more capacity is always useful.
- B Fix the data first: a third engineer with no clean data spends six months waiting, and the two already there are underused.
- C Buy a vendor model instead.
- D Cancel the project.

**Hint.** Ask what the new person would work on during their first month.

**In the session: type the letter in the Zoom chat.**

In the drive-thru trial, McDonald's worked with IBM from 2021, tested at about 100 restaurants, reported accuracy in the low 80s against a target above 95 percent, and ended the arrangement in 2024 while keeping the supplier for other work.

**How should that trial be judged?**

- A The partnership failed, so partnering is the wrong path in general.
- B It worked as intended: a bounded test produced an answer at a known cost, and the decision followed the measurement.
- C They should have built it in-house from the start.
- D They should have bought an off-the-shelf product.

**Hint.** Ask what a limited trial is for, and whether this one answered its question.

**In the session: type the letter in the Zoom chat.**

## Diagnose the two pillars

### What the numbers are

The three positions in the video are here as settings: strong data with no team, strong team with weak data, and both strong with underwriting as the competitive edge. Months to value are the video's own rough figures.

### What to move

Set each pillar and say whether the use case is your competitive edge. The recommended path and the reason are written out underneath. Controls: Data infrastructure; Team capability; Is this use case your competitive edge?.

### Try, in this order

1. Set solid data and no specialists, which is the first insurer, and read the path.
2. Now set patchy data with a strong team, which is the second insurer. Raise the team again and watch the recommendation refuse to move.
3. Raise the data instead, and find the point where the answer changes.

**The point.** The recommendation follows the weaker pillar. Improving the one that is already strong changes nothing.

**1** Diagnose which pillar is weak before choosing build, buy or partner.

**2** Improving the strong pillar does not move the project forward.

**3** Write each gap with a name, a timeline and an owner, or it is not a plan.

# Session 6 is the capstone workshop, Saturday 26 September 2026

## What to do before Saturday

- 1. Take Quiz 6 (Modules 11 and 12), 15 minutes, before Session 6.
- 2. Watch videos V58, V59 and V60 before Saturday. They are short and they are the whole capstone.
- 3. Choose your capstone business problem and bring it written down in one sentence.
- 4. Bring the AI Solution Evaluation Matrix you submitted before today; Session 6 builds directly on it.

Already behind you: the midterm exam on Modules 1 to 6, and the Module 11 evaluation matrix due before today.

## What is assessed, and when

| Item                         | Date                                    | Weight       |
|------------------------------|-----------------------------------------|--------------|
| Quiz 6 (Modules 11 and 12)   | Before 26 September 2026                | Part of 10%  |
| Attendance and participation | Recorded at each live session           | 10%          |
| Capstone brief, first draft  | Thursday 1 October 2026                 | Part of 50%  |
| Two peer reviews written     | Sunday 4 October 2026                   | Part of 50%  |
| Revision after feedback      | Wednesday 8 October 2026                | Part of 50%  |
| Final brief submitted        | Saturday 10 October 2026, hard deadline | 50% in total |